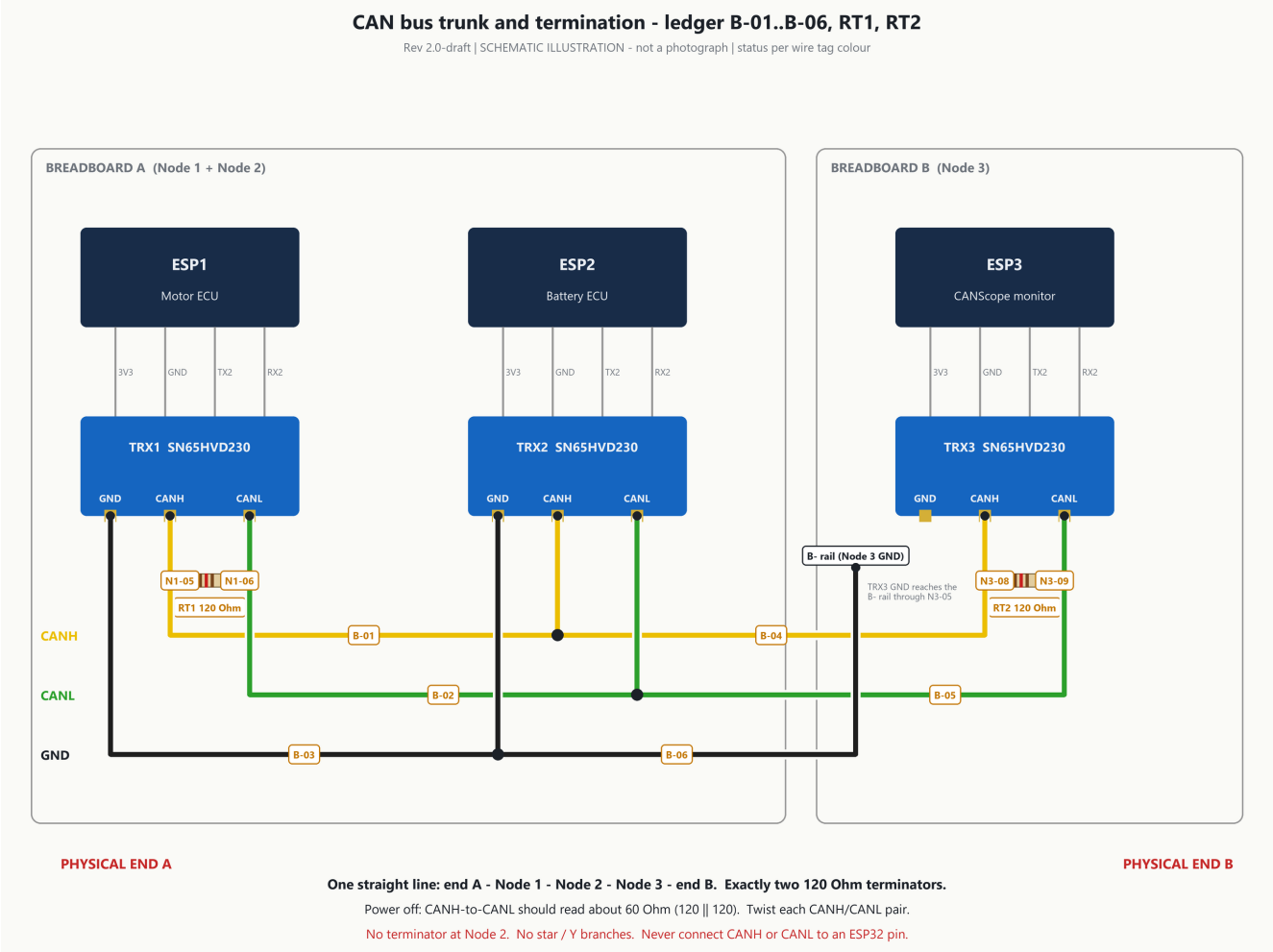
# CANScope

## Three-node ESP32 + Classical CAN bench: physical build manual

**Rev 2.0-draft** | 2026-10-16 | supersedes Rev 1.0 | generated from `docs/wiring/pin_ledger.json`




**NOT FINAL.** This revision corrects Rev 1.0 and assigns every GPIO, but no board, module or wire has been physically inspected, measured or powered by its author. Every diagram is a schematic illustration, not a photograph of your parts. PROVISIONAL connections may be made only after the bench gates listed for them pass. BLOCKED connections (the buzzer transistor Q1) must stay empty. A successful compile never proves the hardware works.

# 0. How to use this manual

Every wire has one row in the canonical ledger (`docs/wiring/pin_ledger.json`). The diagrams, the video, this PDF and the sign-off checklist are generated from it, and `tools/hwdocs/validate.py` checks that they agree with each other and with the firmware pin settings. If anything disagrees with the ledger, the ledger wins. Report the mismatch.

| Status | Meaning | What you do |
|---|---|---|
| **CONFIRMED** | Supported by a named document and independent of unidentified parts | Wire it; check labels |
| **PROVISIONAL** | Supported by documents but depends on a detail only you can see | Pass the listed gates first |
| **BLOCKED** | Required information is missing | Leave it EMPTY; supply the evidence |

Gates G1-G14 are short bench checks. Record each result on the sign-off sheet (section 19).

## Corrections to Rev 1.0

The full register, with evidence, is in `validation/correction_register.md` (27 entries). The most important:

- Node 3 CAN moves to GPIO17/16 (the same as Nodes 1/2 and the repo's monitor firmware). The brief's 22/21 collided with the Arduino-ESP32 default I2C pins SDA 21 / SCL 22, which the OLED needs.

- The SN65HVD230 has no STB/EN pin. Pin 8 is Rs, which must be low for high-speed mode; high means standby (cannot transmit or ACK). Vref stays open.

- All Node 3 peripheral GPIOs assigned: LEDs 25/26/27, buttons 32/33, buzzer drive 13. None are strapping, flash, input-only or USB-serial pins.

- Buttons: diagonal-leg rule replaces the package-dependent 'pairs across the trench' advice.

- Buzzer: reviewed low-side NPN with values; 5 V from Node 3's own VIN (no external supply); transistor pinout BLOCKED until the part is identified.

- Onboard terminators: measured per module (G6) with a decision rule; exactly two on the bus.

- The 3V3 of different nodes is never joined; Board A rails carry GND only.

- Bare boards are flashed before wiring; the bring-up firmware for each node is named.

- Firmware gaps stated plainly: no OLED/LED/button/buzzer code, no simulation, no fault injection yet.

# 1. Inventory and tools

| Qty | Item | Status | Notes |
|---|---|---|---|
| 3 | ESP32 DevKit V1, 30-pin, ESP-WROOM-32 | available | G1 shield marking, G2 header order |
| 3 | SN65HVD230 CAN transceiver module | available | G4 identity, G5 Rs pin, G6 onboard terminator |
| 1 | SSD1306 0.96 in I2C OLED 128x64, 4-pin | available | G7 label order and address |
| 1 each | Green, yellow, red LED | available | G9 polarity |
| 3 | 220 Ohm resistor (one per LED) | available | G13 measure |
| 2 | 120 Ohm resistor (bus ends) | available | G13; fit only per G6 |
| 2 | Tactile push button, 4-leg | available | G8 diagonal legs |
| 1 | 5 V active buzzer | available | G11 type, rating, polarity |
| 2 | 830-point breadboard | available | G14 rails |
| many | Dupont jumper wires | available | Colours suggested in the ledger |
| 3 | USB data cable | available | Charge-only cables will not flash |
| 1 | NPN transistor, Ic >= 100 mA (e.g. PN2222A, 2N3904, BC547, S8050) | REQUIRED EXTRA | Needed so no GPIO drives the buzzer; pinout gate G10 |
| 1 | 1 kOhm resistor (Q1 base) | REQUIRED EXTRA | G13 |
| 1 | 10 kOhm resistor (Q1 base pull-down) | RECOMMENDED | Keeps the buzzer silent during boot |
| 1 | 1N4148 diode | CONDITIONAL | Only if the buzzer is electromagnetic (G11) |
| 1 | Digital multimeter (continuity, diode, Ohm, DC V) | REQUIRED TOOL | Every gate uses it |

# 2. Identification guide

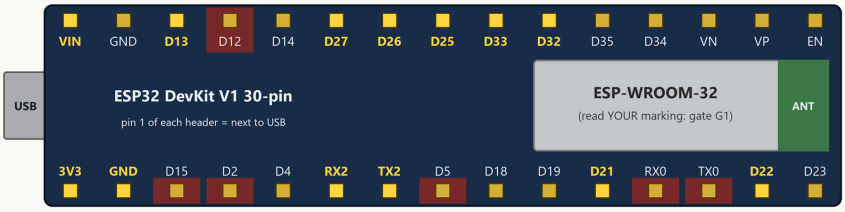
### ESP32 DevKit V1 (30-pin) pin locator: USB to the LEFT, component side up

Highlighted = used by CANScope. Dark red = do NOT use (flash, strapping, USB serial).

**VIN-side header (pin 1 = next to USB)**

| | | | |
|---|---|---|---|
| **1** | **VIN** | **power** | **used 3:BZ1** |
| 2 | GND | power | |
| **3** | **D13** | **GPIO13** | **used 3:R_B** |
| 4 | D12 | GPIO12 | |
| 5 | D14 | GPIO14 | |
| **6** | **D27** | **GPIO27** | **used 3:R_R** |
| **7** | **D26** | **GPIO26** | **used 3:R_Y** |
| **8** | **D25** | **GPIO25** | **used 3:R_G** |

| | | | |
|---|---|---|---|
| **9** | **D33** | **GPIO33** | **used 3:BTN_B** |
| **10** | **D32** | **GPIO32** | **used 3:BTN_A** |
| 11 | D35 | GPIO35 | |
| 12 | D34 | GPIO34 | |
| 13 | VN | GPIO39 | |
| 14 | VP | GPIO36 | |
| 15 | EN | power | |



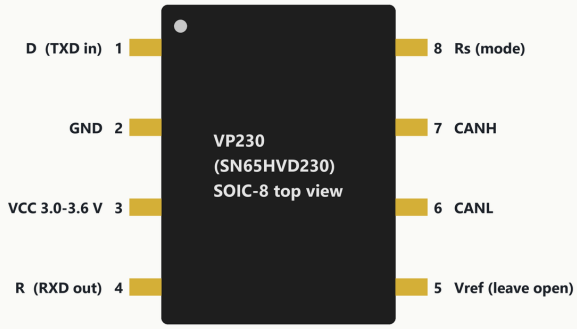

**3V3-side header (pin 1 = next to USB)**

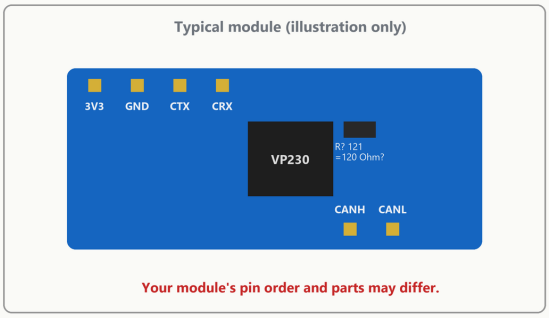
| | | | |
|---|---|---|---|
| **1** | **3V3** | **power** | **used 1:TRX1, 2:TRX2, 3:RAIL3** |
| **2** | **GND** | **power** | **used 1:TRX1, 2:TRX2, 3:RAIL3** |
| 3 | D15 | GPIO15 | |
| 4 | D2 | GPIO2 | |
| 5 | D4 | GPIO4 | |
| **6** | **RX2** | **GPIO16** | **used 1:TRX1, 2:TRX2, 3:TRX3** |
| **7** | **TX2** | **GPIO17** | **used 1:TRX1, 2:TRX2, 3:TRX3** |
| 8 | D5 | GPIO5 | |

| | | | |
|---|---|---|---|
| 9 | D18 | GPIO18 | |
| 10 | D19 | GPIO19 | |
| **11** | **D21** | **GPIO21** | **used 3:OLED** |
| 12 | RX0 | GPIO3 | |
| 13 | TX0 | GPIO1 | |
| **14** | **D22** | **GPIO22** | **used 3:OLED** |
| 15 | D23 | GPIO23 | |

**Gate G2: your silkscreen must match this order exactly, or STOP.**

## SN65HVD230 transceiver: what the IC is, and what to check on YOUR module



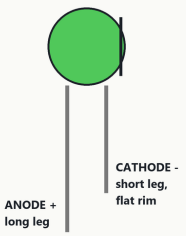




| | |
|---|---|
| **Supply** | VCC 3.0-3.6 V. Feed it from the ESP32 3V3 pin. Never 5 V (D/R abs max = VCC+0.5 V). |
| **TX path** | ESP32 TX GPIO -> module CTX / TXD / D. NOT crossed like a UART. |
| **RX path** | Module CRX / RXD / R -> ESP32 RX GPIO. |
| **Rs pin 8** | Strong pull-down = high-speed (needed). 10-100 kOhm = slope control. High = STANDBY: driver off, cannot transmit or ACK. Gate G5. |
| **Vref pin 5** | VCC/2 reference output. Leave unconnected. |
| **Termination** | Many modules carry a 120 Ohm across CANH-CANL. Measure it (gate G6). |
| **Labels** | Module header order varies. Wire by PRINTED NAME only (gate G4). |

Source: Texas Instruments SN65HVD23x datasheet SLOS346O (rev. April 2018), pin table and Section 8.

## Identify before you insert: LED, resistor, button, buzzer, transistor, OLED

### LED polarity (gate G9)



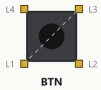

GPIO -> 220 Ohm -> anode; cathode -> GND

### Resistor colour bands (4-band) - measure anyway (G13)



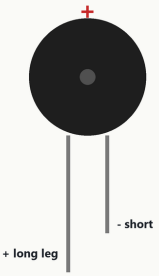

5-band (1%) parts use different colours.

### Tactile button: diagonal rule (G8)



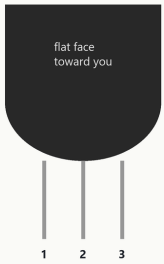

Diagonal legs (L1, L3) are ALWAYS on opposite contacts, in any orientation.
GPIO -> L1, GND -> L3. Meter: open released, closed pressed.

### 5 V active buzzer (G11)




Active = beeps on plain DC.
Rated 5 V, <= 30 mA. Never on a GPIO.
Electromagnetic type -> add D1.

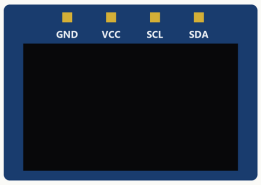
### NPN transistor TO-92 (G10) - BLOCKED until read




```
PN2222A   legs 1-2-3 = E B C
2N3904    legs 1-2-3 = E B C
S8050     legs 1-2-3 = E B C
BC547     legs 1-2-3 = C B E
```

**Examples only: confirm with YOUR part's datasheet.**

### SSD1306 I2C OLED (G7)



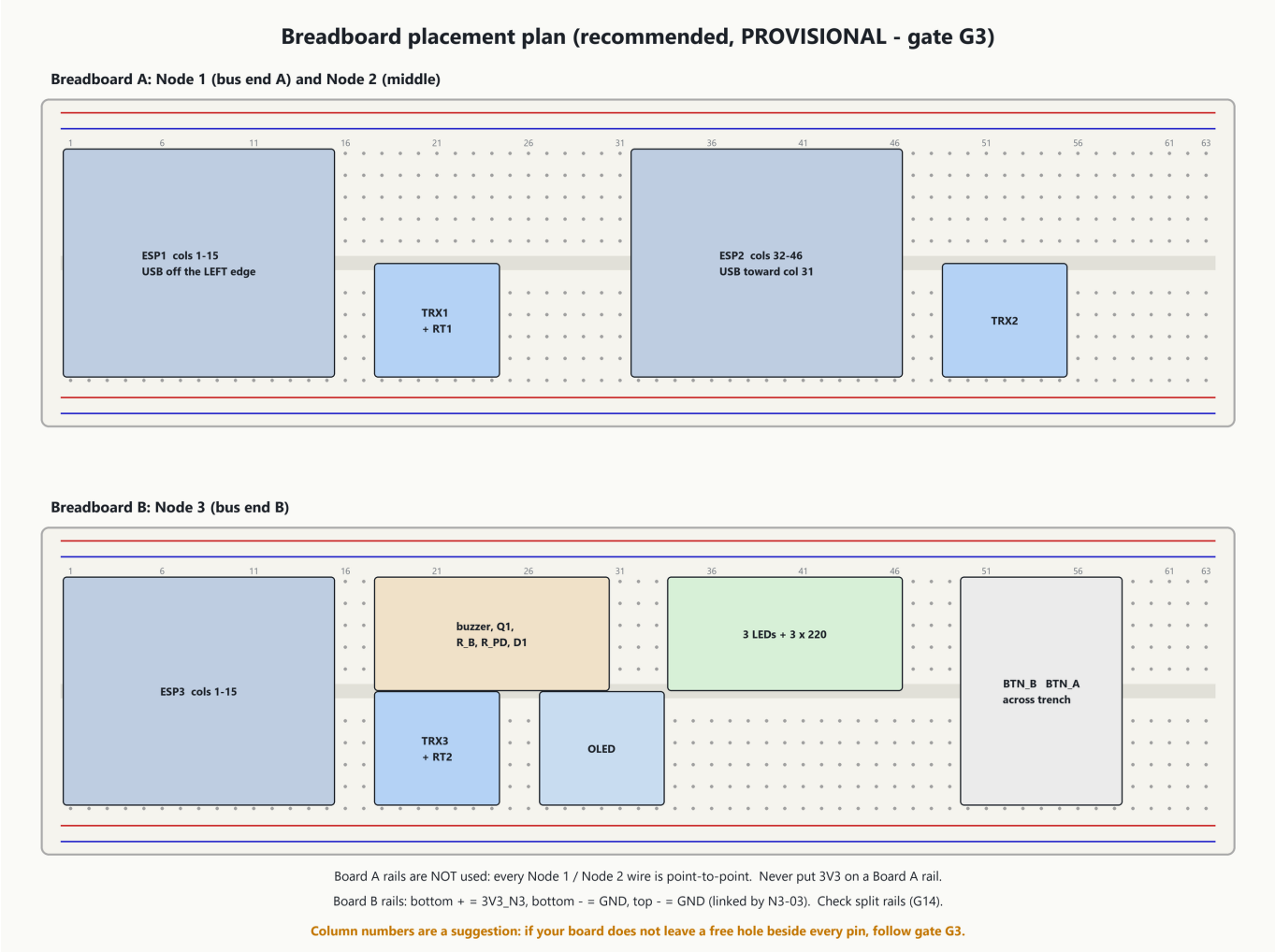

Read YOUR labels: some modules swap GND and VCC.
Power from 3V3 only (keeps SDA/SCL pull-ups at 3.3 V).
Address: back marking 0x78 = 0x3C.

## Bench gates

| Gate | Title | When | Procedure | Pass | Fail |
|---|---|---|---|---|---|
| G1 | ESP32 module identity | Before any wire | Read the laser marking on the metal shield of each ESP32. Photograph it. | It reads ESP32-WROOM-32 (or WROOM-32D/32E/32U). Not WROVER, not ESP32-S2/S3/C3. | STOP. GPIO16/17 may be PSRAM pins. Send the photo; the ledger must be revised. |
| G2 | Header silkscreen matches the ledger | Before any wire | With USB pointing left, read every pin label on both headers and compare with the 30-pin table (3V3 side: 3V3, GND, D15, D2, D4, RX2, TX2, D5, D18, D19, D21, RX0, TX0, D22, D23; VIN side: VIN, GND, D13, D12, D14, D27, D26, D25, D33, D32, D35, D34, VN, VP, EN). | All 30 labels match in the same order. | STOP. Do not count pins from the diagram. Send a photo of both headers. |
| G3 | Breadboard fit | When seating each ESP32 | Seat the ESP32 across the centre trench. Count free holes beside each pin row. | At least one free hole remains in the same 5-hole strip as every pin, on both sides. | Do not force it. Bridge the board across two breadboards (one header row in each) or use female-to-male jumpers on the header pins. Record which. |
| G4 | Transceiver module identity and labels | Before wiring a transceiver | Read the 8-pin IC marking (expect VP230) and every printed pin label. Check whether the module has any 5 V pin or a voltage regulator. Photograph front and back. | IC reads VP230; supply pin is labelled 3V3/3.3V/VCC with no 5 V-only marking; you can find pins equivalent to 3V3, GND, CTX (TXD/D), CRX (RXD/R), CANH, CANL. | STOP. Different IC or a 5 V module needs a new design review. |
| G5 | Transceiver Rs (mode) pin | After G4 | If the module exposes a pin labelled RS, S or STB: it must be wired to GND (see the conditional rows). If not exposed: with all power removed, measure resistance from IC pin 8 (Rs) to IC pin 2 (GND). | Exposed RS pin wired to GND, OR measured Rs-GND below 100 Ohm (high-speed mode per TI SLOS346O). | Rs-GND 10-100 kOhm (slope control) or open/high (standby, cannot transmit or ACK): STOP and record the value. |
| G6 | Onboard termination | Before installing RT1/RT2 | With the module unpowered and NOT connected to anything, measure CANH to CANL on each transceiver module. | Reading above 10 kOhm (no onboard terminator) on all three modules: fit RT1 and RT2 exactly as the ledger says. | About 120 Ohm on a module: it has an onboard terminator. On TRX1/TRX3 omit the external RT1/RT2 at that node. On TRX2 (middle node): STOP, the onboard resistor must be removed or its jumper opened; send a photo first. |
| G7 | OLED labels | Before wiring the OLED | Read the four header labels left to right and the address marking on the back (0x78 = 0x3C, 0x7A = 0x3D). | Labels GND, VCC (or VDD), SCL (or SCK), SDA are present; record their order and address. | Different labels (e.g. SPI pins CS/DC/RES): STOP, it is not an I2C module. |
| G8 | Push-button contacts | Before inserting each button | Continuity mode: probe two diagonally opposite legs. Released: open. Pressed: beep / near 0 Ohm. | Diagonal legs are open when released and closed when pressed. | Rotate the meter probes to find a released-open / pressed-closed leg pair; record it and use it instead of the diagonal pair. |
| G9 | LED polarity | Before inserting each LED | Long leg = anode (+). Flat edge on the rim = cathode (-). Confirm with the meter diode-test mode (red probe on anode lights the LED faintly). | Anode and cathode identified for every LED. | Legs cut equal and no flat edge visible: use diode test only. |
| G10 | Transistor part and pinout | Before inserting Q1 | Read the transistor marking. Look up E/B/C in that exact part's manufacturer datasheet. Common TO-92 examples (flat face toward you, legs down, left to right): PN2222A E-B-C; 2N3904 E-B-C; S8050 E-B-C; BC547 C-B-E. Your part may differ. | Part number and E/B/C order recorded from its own datasheet; Ic(max) >= 100 mA; NPN. | Unknown marking or PNP: STOP. |
| G11 | Buzzer identity | Before wiring the buzzer | Confirm it is ACTIVE (sounds on plain DC), rated 5 V, current <= 30 mA, and find + (marked or longer leg). Note whether it is electromagnetic (needs flyback diode) or piezo. | Active 5 V, <= 30 mA, + identified, type recorded. | Passive buzzer, > 30 mA, or no polarity mark: STOP for design review. |
| G12 | Node 3 VIN voltage | First power-up of Node 3, buzzer + wire NOT installed | Node 3 on USB only. Measure VIN to GND in DC volts. | Between 4.4 V and 5.25 V. | Outside that range: do not install the buzzer supply wire; record value. |

| Gate | Title | When | Procedure | Pass | Fail |
|---|---|---|---|---|---|
| G13 | Resistor values | Before inserting each resistor | Measure every resistor out of circuit. | 220 Ohm: 209-231; 120 Ohm: 114-126; 1 kOhm: 950-1050; 10 kOhm: 9.5k-10.5k. | Wrong value: set it aside. |
| G14 | Breadboard rails | Before using a rail | Continuity from one end of each rail to the other. | Continuous end to end, or the split point is marked. | Split rail: add a short bridge jumper on the same rail only, or keep loads on one half. |

# 3. Breadboard layout and node labelling




Label each board with tape before placing: N1 Motor ECU, N2 Battery ECU, N3 CANScope. Board A holds Nodes 1 and 2; Board B holds Node 3. The column numbers are suggestions; gate G3 decides the fit.

Before any wire: flash each **bare** board (nothing connected but USB), one at a time. Node 1: `pio run -d firmware/node1 -e node1 -t upload`. Node 2: `pio run -d firmware/node2 -e legacy_sketch -t upload`. Node 3: `pio run -d firmware/monitor -e monitor -t upload`. All use CAN TX GPIO17, RX GPIO16, 500 kbit/s, which matches the ledger (checked by validate.py).

# 4. Pin-by-pin wiring ledger (all connections)

ESP32 positions count from the USB end of each header, as in the pin locator. 'Lead' means a component's own leg sharing a breadboard strip; no jumper. Colours are suggestions; the ID label on the wire is authoritative.

| ID | From | To | Signal | Wire | Status | Gates | Check (power off) |
|---|---|---|---|---|---|---|---|
| N1-01 | ESP1 3V3, 3V3-side pin 1 from USB | TRX1 3V3 | 3V3_N1 | red | PROVISIONAL | G1, G2, G4 | Power off: continuity ESP1 3V3 pin to TRX1 3V3 pin; no continuity to any GND or 5 V point. |
| N1-02 | ESP1 GND, 3V3-side pin 2 from USB | TRX1 GND | GND | black | PROVISIONAL | G2, G4 | Continuity ESP1 GND to TRX1 GND. |
| N1-03 | ESP1 TX2 (GPIO17), 3V3-side pin 7 from USB | TRX1 CTX | CAN_TX_N1 (GPIO17) | white | PROVISIONAL | G1, G2, G4 | Continuity TX2 pin to CTX pin; NOT to CRX. |
| N1-04 | ESP1 RX2 (GPIO16), 3V3-side pin 6 from USB | TRX1 CRX | CAN_RX_N1 (GPIO16) | brown | PROVISIONAL | G1, G2, G4 | Continuity RX2 pin to CRX pin; NOT to CTX. |
| N1-05 | RT1 1 | TRX1 CANH | CANH (end A terminator) | lead | PROVISIONAL | G6, G13 | Fit only if G6 found no onboard terminator on TRX1. |
| N1-06 | RT1 2 | TRX1 CANL | CANL (end A terminator) | lead | PROVISIONAL | G6, G13 | RT1 measured 114-126 Ohm before fitting. |
| N2-01 | ESP2 3V3, 3V3-side pin 1 from USB | TRX2 3V3 | 3V3_N2 | red | PROVISIONAL | G1, G2, G4 | Continuity ESP2 3V3 to TRX2 3V3. NO continuity to Node 1 3V3. |
| N2-02 | ESP2 GND, 3V3-side pin 2 from USB | TRX2 GND | GND | black | PROVISIONAL | G2, G4 | Continuity ESP2 GND to TRX2 GND. |
| N2-03 | ESP2 TX2 (GPIO17), 3V3-side pin 7 from USB | TRX2 CTX | CAN_TX_N2 (GPIO17) | white | PROVISIONAL | G1, G2, G4 | Continuity TX2 pin to CTX pin; NOT to CRX. |
| N2-04 | ESP2 RX2 (GPIO16), 3V3-side pin 6 from USB | TRX2 CRX | CAN_RX_N2 (GPIO16) | brown | PROVISIONAL | G1, G2, G4 | Continuity RX2 pin to CRX pin; NOT to CTX. |
| N3-01 | ESP3 3V3, 3V3-side pin 1 from USB | Board B bottom + rail (3V3_N3) | 3V3_N3 | red | PROVISIONAL | G2, G14 | Continuity 3V3 pin to bottom + rail. |
| N3-02 | ESP3 GND, 3V3-side pin 2 from USB | Board B bottom - rail (GND) | GND | black | PROVISIONAL | G2, G14 | Continuity GND pin (3V3 side) to bottom - rail. |
| N3-03 | Board B bottom - rail (GND) | Board B top - rail (GND) | GND (rail link, far right end of board B) | black | PROVISIONAL | G14 | Continuity bottom - rail to top - rail: near 0 Ohm. Never link a - rail to a + rail. |
| N3-04 | Board B bottom + rail (3V3_N3) | TRX3 3V3 | 3V3_N3 | red | PROVISIONAL | G4 | Continuity + rail to TRX3 3V3. |
| N3-05 | Board B bottom - rail (GND) | TRX3 GND | GND | black | PROVISIONAL | G4 | Continuity - rail to TRX3 GND. |
| N3-06 | ESP3 TX2 (GPIO17), 3V3-side pin 7 from USB | TRX3 CTX | CAN_TX_N3 (GPIO17) | white | PROVISIONAL | G1, G2, G4 | Continuity TX2 to CTX; NOT to CRX. |
| N3-07 | ESP3 RX2 (GPIO16), 3V3-side pin 6 from USB | TRX3 CRX | CAN_RX_N3 (GPIO16) | brown | PROVISIONAL | G1, G2, G4 | Continuity RX2 to CRX; NOT to CTX. |
| N3-08 | RT2 1 | TRX3 CANH | CANH (end B terminator) | lead | PROVISIONAL | G6, G13 | Fit only if G6 found no onboard terminator on TRX3. |
| N3-09 | RT2 2 | TRX3 CANL | CANL (end B terminator) | lead | PROVISIONAL | G6, G13 | RT2 measured 114-126 Ohm before fitting. |
| N3-10 | Board B bottom + rail (3V3_N3) | OLED VCC | 3V3_N3 | red | PROVISIONAL | G7 | Continuity + rail to the pin PRINTED VCC/VDD (not by position). |
| N3-11 | Board B bottom - rail (GND) | OLED GND | GND | black | PROVISIONAL | G7 | Continuity - rail to the pin PRINTED GND. |
| N3-12 | ESP3 D21 (GPIO21), 3V3-side pin 11 from USB | OLED SDA | I2C_SDA (GPIO21) | blue | PROVISIONAL | G2, G7 | Continuity D21 to SDA; NOT to SCL. |
| N3-13 | ESP3 D22 (GPIO22), 3V3-side pin 14 from USB | OLED SCL | I2C_SCL (GPIO22) | purple | PROVISIONAL | G2, G7 | Continuity D22 to SCL; NOT to SDA. |
| N3-14 | ESP3 D25 (GPIO25), VIN-side pin 8 from USB | R_G 1 | LED_GREEN (GPIO25) | grey | PROVISIONAL | G2, G13 | Continuity D25 to one end of the green 220 Ohm. |

| ID | From | To | Signal | Wire | Status | Gates | Check (power off) |
|---|---|---|---|---|---|---|---|
| N3-15 | R_G 2 | LED_G A | LED_GREEN via 220 | lead | PROVISIONAL | G9 | Resistor and anode share one 5-hole strip; nothing else in it. |
| N3-16 | LED_G K | Board B top - rail (GND) | GND | lead | PROVISIONAL | G9, G14 | Cathode (short leg, flat side) to top GND rail. |
| N3-17 | ESP3 D26 (GPIO26), VIN-side pin 7 from USB | R_Y 1 | LED_YELLOW (GPIO26) | grey | PROVISIONAL | G2, G13 | Continuity D26 to one end of the yellow 220 Ohm. |
| N3-18 | R_Y 2 | LED_Y A | LED_YELLOW via 220 | lead | PROVISIONAL | G9 | Resistor and anode share one strip. |
| N3-19 | LED_Y K | Board B top - rail (GND) | GND | lead | PROVISIONAL | G9, G14 | Cathode to top GND rail. |
| N3-20 | ESP3 D27 (GPIO27), VIN-side pin 6 from USB | R_R 1 | LED_RED (GPIO27) | grey | PROVISIONAL | G2, G13 | Continuity D27 to one end of the red 220 Ohm. |
| N3-21 | R_R 2 | LED_R A | LED_RED via 220 | lead | PROVISIONAL | G9 | Resistor and anode share one strip. |
| N3-22 | LED_R K | Board B top - rail (GND) | GND | lead | PROVISIONAL | G9, G14 | Cathode to top GND rail. |
| N3-23 | ESP3 D32 (GPIO32), VIN-side pin 10 from USB | BTN_A L1 | BUTTON_A (GPIO32, internal pull-up, active LOW) | white | PROVISIONAL | G2, G8 | Continuity D32 to leg L1 of button A. |
| N3-24 | BTN_A L3 | Board B top - rail (GND) | GND | black | PROVISIONAL | G8 | L3 is DIAGONAL to L1. D32-to-GND: open released, ~0 Ohm pressed. |
| N3-25 | ESP3 D33 (GPIO33), VIN-side pin 9 from USB | BTN_B L1 | BUTTON_B (GPIO33, internal pull-up, active LOW) | white | PROVISIONAL | G2, G8 | Continuity D33 to leg L1 of button B. |
| N3-26 | BTN_B L3 | Board B top - rail (GND) | GND | black | PROVISIONAL | G8 | D33-to-GND: open released, ~0 Ohm pressed. |
| N3-27 | ESP3 D13 (GPIO13), VIN-side pin 3 from USB | R_B 1 | BUZZER_DRIVE (GPIO13) | grey | PROVISIONAL | G2, G13 | Continuity D13 (3rd pin from USB, VIN side) to one end of the 1 kOhm. The 4th pin, D12, is a strapping pin: make sure it stays EMPTY. D13 has NO path to the buzzer except through Q1. |
| N3-28 | R_B 2 | Q1 B | Q1 base via 1 kOhm | lead | BLOCKED | G10 | Blocked until Q1 part number and its E/B/C order are recorded. |
| N3-29 | R_PD 1 | Q1 B | base pull-down (boot-quiet) | lead | BLOCKED | G10 | Blocked until Q1 pinout is recorded. |
| N3-30 | R_PD 2 | Board B top - rail (GND) | GND | lead | BLOCKED | G10 | Blocked until Q1 pinout is recorded. |
| N3-31 | Q1 E | Board B top - rail (GND) | GND (emitter) | black | BLOCKED | G10 | Blocked until Q1 pinout is recorded. |
| N3-32 | Q1 C | BZ1 - | BUZZER_LOW (collector) | lead | BLOCKED | G10, G11 | Blocked until Q1 pinout and buzzer polarity are recorded. |
| N3-33 | D1 A | BZ1 - | flyback diode anode | lead | PROVISIONAL | G11 | Only if G11 says electromagnetic. Band (cathode) goes to the + side. |
| N3-34 | D1 K | BZ1 + | flyback diode cathode | lead | PROVISIONAL | G11 | Band toward buzzer +. |
| N3-35 | ESP3 VIN, VIN-side pin 1 from USB | BZ1 + | 5V_USB_N3 (buzzer supply) | orange | PROVISIONAL | G11, G12 | Install LAST, with power off, after G12. No continuity from this wire to any 3V3 pin or rail. |
| B-01 | TRX1 CANH | TRX2 CANH | CANH trunk, segment 1 | yellow | PROVISIONAL | G4 | Continuity TRX1 CANH to TRX2 CANH; no continuity CANH to CANL other than through terminators. |
| B-02 | TRX1 CANL | TRX2 CANL | CANL trunk, segment 1 | green | PROVISIONAL | G4 | Twist with B-01. Continuity TRX1 CANL to TRX2 CANL. |
| B-03 | TRX1 GND | TRX2 GND | GND reference, segment 1 | black | PROVISIONAL | G4 | Continuity Node 1 GND to Node 2 GND. |
| B-04 | TRX2 CANH | TRX3 CANH | CANH trunk, segment 2 (board A to B) | yellow | PROVISIONAL | G4 | Continuity TRX2 CANH to TRX3 CANH. |
| B-05 | TRX2 CANL | TRX3 CANL | CANL trunk, segment 2 (board A to B) | green | PROVISIONAL | G4 | Twist with B-04. Continuity TRX2 CANL to TRX3 CANL. |

| ID | From | To | Signal | Wire | Status | Gates | Check (power off) |
|---|---|---|---|---|---|---|---|
| B-06 | TRX2 GND | Board B bottom - rail (GND) | GND reference, segment 2 | black | PROVISIONAL | G4, G14 | Continuity Node 2 GND to Node 3 GND rail. |

## Conditional rows

| ID | Applies when | From | To | Status | Gates |
|---|---|---|---|---|---|
| C-RS1 | Only if TRX1 exposes a pin labelled RS, S or STB | TRX1:RS | TRX1:GND | PROVISIONAL | G4, G5 |
| C-RS2 | Only if TRX2 exposes a pin labelled RS, S or STB | TRX2:RS | TRX2:GND | PROVISIONAL | G4, G5 |
| C-RS3 | Only if TRX3 exposes a pin labelled RS, S or STB | TRX3:RS | RAIL3:GND- | PROVISIONAL | G4, G5 |
| C-VREF | Any module exposing Vref | TRXn:Vref | (no connection) | CONFIRMED | |
| C-T2 | If G6 finds an onboard 120 Ohm on TRX2 (middle node) | TRX2 onboard terminator | (remove / open jumper) | BLOCKED | G6 |

# 5. Node 1 - Motor ECU wiring

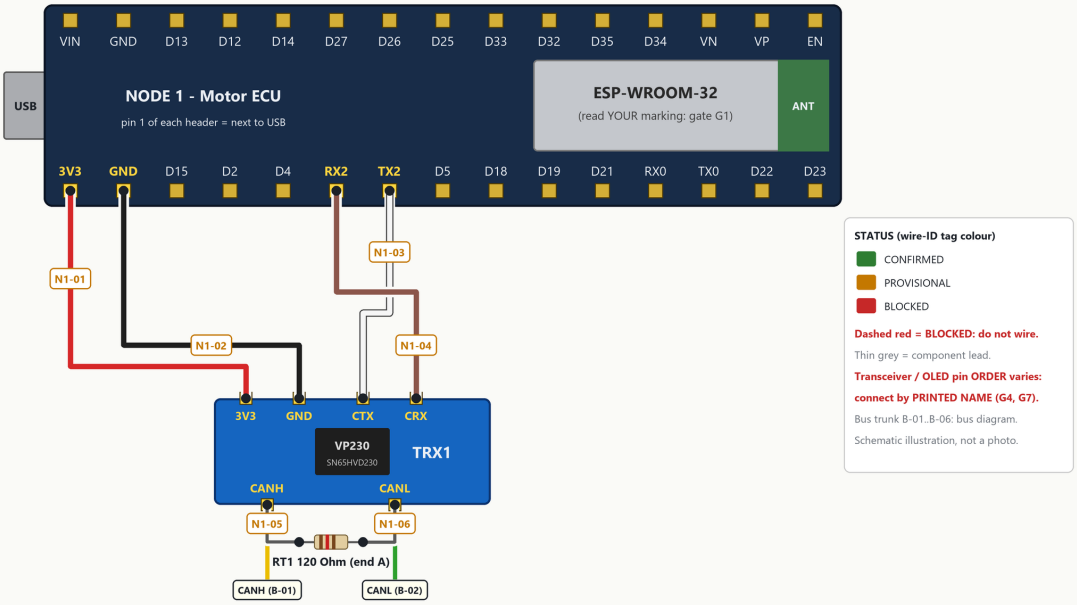
**Node 1 (Motor ECU) wiring - ledger N1-01..N1-06**

Rev 2.0-draft | SCHEMATIC ILLUSTRATION - not a photograph | status per wire tag colour

**Node 1 uses only FOUR ESP32 pins, all on the 3V3-side header (bottom row):**

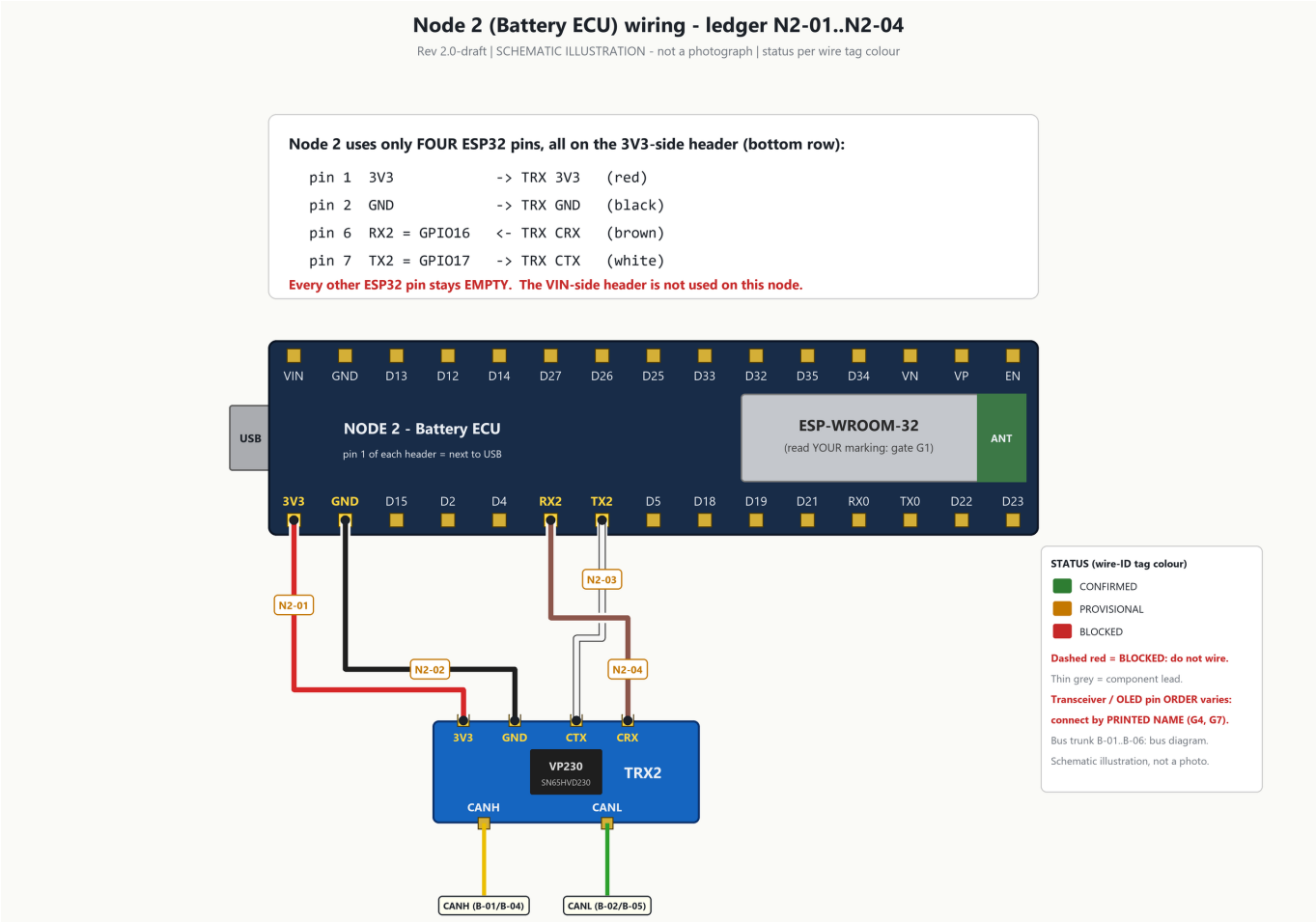
```
pin 1  3V3            -> TRX 3V3   (red)
pin 2  GND            -> TRX GND   (black)
pin 6  RX2 = GPIO16   <- TRX CRX   (brown)
pin 7  TX2 = GPIO17   -> TRX CTX   (white)
```

**Every other ESP32 pin stays EMPTY. The VIN-side header is not used on this node.**




| ID | From | To | Signal | Wire | Status | Gates | Check (power off) |
|---|---|---|---|---|---|---|---|
| N1-01 | ESP1 3V3, 3V3-side pin 1 from USB | TRX1 3V3 | 3V3_N1 | red | **PROVISIONAL** | G1, G2, G4 | Power off: continuity ESP1 3V3 pin to TRX1 3V3 pin; no continuity to any GND or 5 V point. |
| N1-02 | ESP1 GND, 3V3-side pin 2 from USB | TRX1 GND | GND | black | **PROVISIONAL** | G2, G4 | Continuity ESP1 GND to TRX1 GND. |
| N1-03 | ESP1 TX2 (GPIO17), 3V3-side pin 7 from USB | TRX1 CTX | CAN_TX_N1 (GPIO17) | white | **PROVISIONAL** | G1, G2, G4 | Continuity TX2 pin to CTX pin; NOT to CRX. |
| N1-04 | ESP1 RX2 (GPIO16), 3V3-side pin 6 from USB | TRX1 CRX | CAN_RX_N1 (GPIO16) | brown | **PROVISIONAL** | G1, G2, G4 | Continuity RX2 pin to CRX pin; NOT to CTX. |
| N1-05 | RT1 1 | TRX1 CANH | CANH (end A terminator) | lead | **PROVISIONAL** | G6, G13 | Fit only if G6 found no onboard terminator on TRX1. |
| N1-06 | RT1 2 | TRX1 CANL | CANL (end A terminator) | lead | **PROVISIONAL** | G6, G13 | RT1 measured 114-126 Ohm before fitting. |

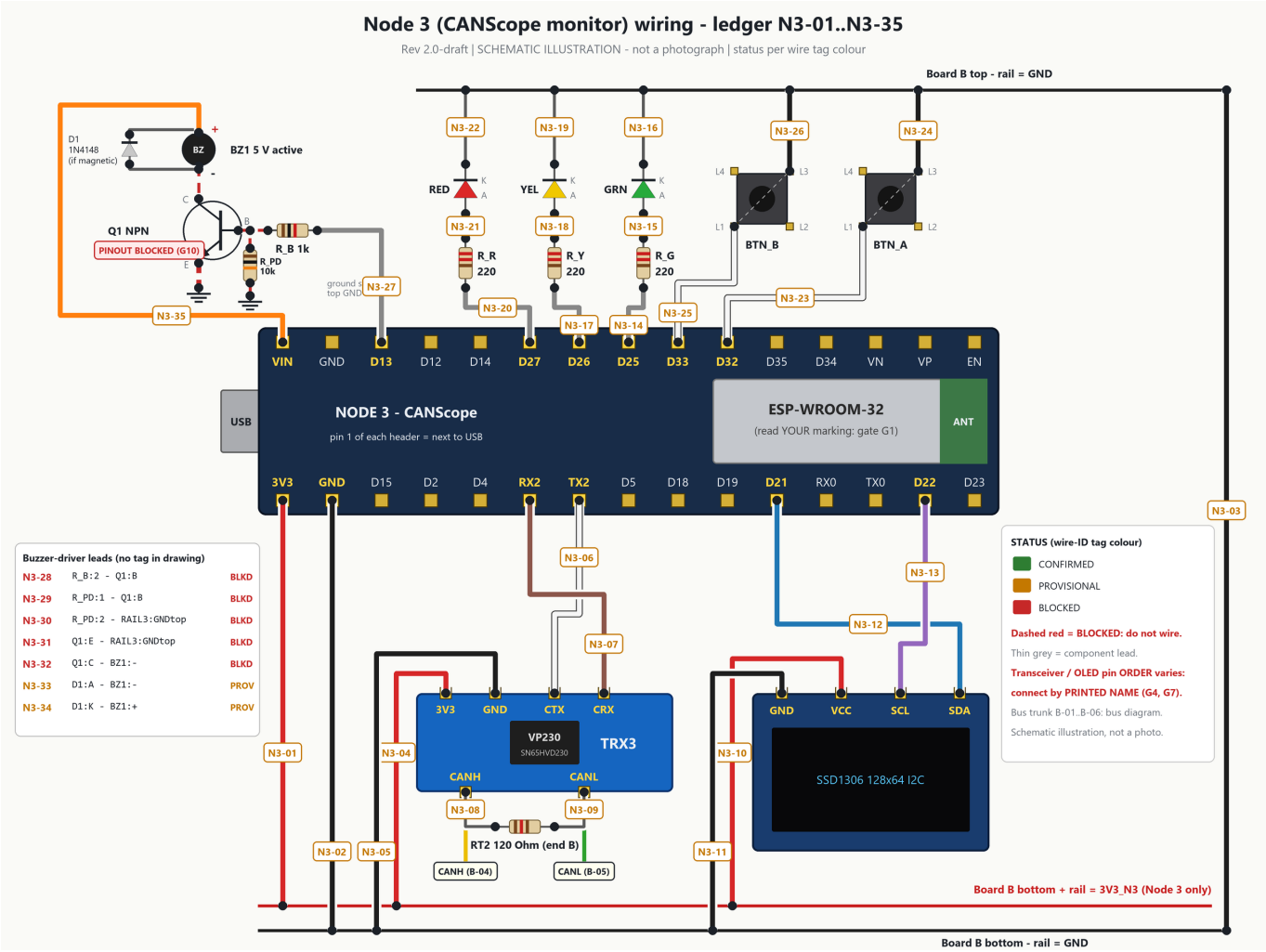
# 6. Node 2 - Battery ECU wiring




| ID | From | To | Signal | Wire | Status | Gates | Check (power off) |
|---|---|---|---|---|---|---|---|
| N2-01 | ESP2 3V3, 3V3-side pin 1 from USB | TRX2 3V3 | 3V3_N2 | red | **PROVISIONAL** | G1, G2, G4 | Continuity ESP2 3V3 to TRX2 3V3. NO continuity to Node 1 3V3. |
| N2-02 | ESP2 GND, 3V3-side pin 2 from USB | TRX2 GND | GND | black | **PROVISIONAL** | G2, G4 | Continuity ESP2 GND to TRX2 GND. |
| N2-03 | ESP2 TX2 (GPIO17), 3V3-side pin 7 from USB | TRX2 CTX | CAN_TX_N2 (GPIO17) | white | **PROVISIONAL** | G1, G2, G4 | Continuity TX2 pin to CTX pin; NOT to CRX. |
| N2-04 | ESP2 RX2 (GPIO16), 3V3-side pin 6 from USB | TRX2 CRX | CAN_RX_N2 (GPIO16) | brown | **PROVISIONAL** | G1, G2, G4 | Continuity RX2 pin to CRX pin; NOT to CTX. |

Shared with Node 1: the same four pins and colours. Node-specific: Node 2 is the middle of the bus (no terminator), and its 3V3 must never touch Node 1's 3V3.

# 7. Node 3 - CANScope monitor wiring




Rows for Node 3 are listed by circuit in sections 8-11; power and transceiver rows below.

| ID | From | To | Signal | Wire | Status | Gates | Check (power off) |
|---|---|---|---|---|---|---|---|
| N3-01 | ESP3 3V3, 3V3-side pin 1 from USB | Board B bottom + rail (3V3_N3) | 3V3_N3 | red | **PROVISIONAL** | G2, G14 | Continuity 3V3 pin to bottom + rail. |
| N3-02 | ESP3 GND, 3V3-side pin 2 from USB | Board B bottom - rail (GND) | GND | black | **PROVISIONAL** | G2, G14 | Continuity GND pin (3V3 side) to bottom - rail. |
| N3-03 | Board B bottom - rail (GND) | Board B top - rail (GND) | GND (rail link, far right end of board B) | black | **PROVISIONAL** | G14 | Continuity bottom - rail to top - rail: near 0 Ohm. Never link a - rail to a + rail. |
| N3-04 | Board B bottom + rail (3V3_N3) | TRX3 3V3 | 3V3_N3 | red | **PROVISIONAL** | G4 | Continuity + rail to TRX3 3V3. |
| N3-05 | Board B bottom - rail (GND) | TRX3 GND | GND | black | **PROVISIONAL** | G4 | Continuity - rail to TRX3 GND. |
| N3-06 | ESP3 TX2 (GPIO17), 3V3-side pin 7 from USB | TRX3 CTX | CAN_TX_N3 (GPIO17) | white | **PROVISIONAL** | G1, G2, G4 | Continuity TX2 to CTX; NOT to CRX. |
| N3-07 | ESP3 RX2 (GPIO16), 3V3-side pin 6 from USB | TRX3 CRX | CAN_RX_N3 (GPIO16) | brown | **PROVISIONAL** | G1, G2, G4 | Continuity RX2 to CRX; NOT to CTX. |

# 8. OLED wiring and orientation

Wire by the printed pin name, never by position: modules exist with GND-VCC-SCL-SDA and VCC-GND-SCL-SDA orders (gate G7). Power from the 3V3 rail only, so the module's I2C pull-ups sit at 3.3 V. SDA = GPIO21 and SCL = GPIO22 are the Arduino-ESP32 default Wire pins (pins_arduino.h). The address is usually 0x3C (back marking 0x78). It can only be confirmed by an I2C scan, and **no firmware for it exists yet**.

| ID | From | To | Signal | Wire | Status | Gates | Check (power off) |
|---|---|---|---|---|---|---|---|
| N3-10 | Board B bottom + rail (3V3_N3) | OLED VCC | 3V3_N3 | red | PROVISIONAL | G7 | Continuity + rail to the pin PRINTED VCC/VDD (not by position). |
| N3-11 | Board B bottom - rail (GND) | OLED GND | GND | black | PROVISIONAL | G7 | Continuity - rail to the pin PRINTED GND. |
| N3-12 | ESP3 D21 (GPIO21), 3V3-side pin 11 from USB | OLED SDA | I2C_SDA (GPIO21) | blue | PROVISIONAL | G2, G7 | Continuity D21 to SDA; NOT to SCL. |
| N3-13 | ESP3 D22 (GPIO22), 3V3-side pin 14 from USB | OLED SCL | I2C_SCL (GPIO22) | purple | PROVISIONAL | G2, G7 | Continuity D22 to SCL; NOT to SDA. |

# 9. Three LED circuits

Each LED has its own 220 Ohm series resistor: GPIO -> resistor -> anode (long leg); cathode (short leg, flat rim) -> GND. Current = (3.3 V - Vf) / 220 Ohm, about 5-7 mA. Never share one resistor between LEDs.

| ID | From | To | Signal | Wire | Status | Gates | Check (power off) |
|---|---|---|---|---|---|---|---|
| N3-14 | ESP3 D25 (GPIO25), VIN-side pin 8 from USB | R_G 1 | LED_GREEN (GPIO25) | grey | PROVISIONAL | G2, G13 | Continuity D25 to one end of the green 220 Ohm. |
| N3-15 | R_G 2 | LED_G A | LED_GREEN via 220 | lead | PROVISIONAL | G9 | Resistor and anode share one 5-hole strip; nothing else in it. |
| N3-16 | LED_G K | Board B top - rail (GND) | GND | lead | PROVISIONAL | G9, G14 | Cathode (short leg, flat side) to top GND rail. |
| N3-17 | ESP3 D26 (GPIO26), VIN-side pin 7 from USB | R_Y 1 | LED_YELLOW (GPIO26) | grey | PROVISIONAL | G2, G13 | Continuity D26 to one end of the yellow 220 Ohm. |
| N3-18 | R_Y 2 | LED_Y A | LED_YELLOW via 220 | lead | PROVISIONAL | G9 | Resistor and anode share one strip. |
| N3-19 | LED_Y K | Board B top - rail (GND) | GND | lead | PROVISIONAL | G9, G14 | Cathode to top GND rail. |
| N3-20 | ESP3 D27 (GPIO27), VIN-side pin 6 from USB | R_R 1 | LED_RED (GPIO27) | grey | PROVISIONAL | G2, G13 | Continuity D27 to one end of the red 220 Ohm. |
| N3-21 | R_R 2 | LED_R A | LED_RED via 220 | lead | PROVISIONAL | G9 | Resistor and anode share one strip. |
| N3-22 | LED_R K | Board B top - rail (GND) | GND | lead | PROVISIONAL | G9, G14 | Cathode to top GND rail. |

# 10. Buttons: contacts and orientation

On a 4-leg tactile switch the two contacts each have two legs, so **diagonally opposite legs are always on different contacts**, however the switch is turned. Seat it across the centre trench, wire GPIO to L1 and GND to the diagonal leg L3, and confirm with the meter: open when released, closed when pressed (G8). The firmware will use the internal pull-up (about 45 kOhm), so pressed = LOW. GPIO32/33 have internal pull-ups; GPIO34-39 do not and are not used.

| ID | From | To | Signal | Wire | Status | Gates | Check (power off) |
|---|---|---|---|---|---|---|---|
| N3-23 | ESP3 D32 (GPIO32), VIN-side pin 10 from USB | BTN_A L1 | BUTTON_A (GPIO32, internal pull-up, active LOW) | white | PROVISIONAL | G2, G8 | Continuity D32 to leg L1 of button A. |
| N3-24 | BTN_A L3 | Board B top - rail (GND) | GND | black | PROVISIONAL | G8 | L3 is DIAGONAL to L1. D32-to-GND: open released, ~0 Ohm pressed. |
| N3-25 | ESP3 D33 (GPIO33), VIN-side pin 9 from USB | BTN_B L1 | BUTTON_B (GPIO33, internal pull-up, active LOW) | white | PROVISIONAL | G2, G8 | Continuity D33 to leg L1 of button B. |
| N3-26 | BTN_B L3 | Board B top - rail (GND) | GND | black | PROVISIONAL | G8 | D33-to-GND: open released, ~0 Ohm pressed. |

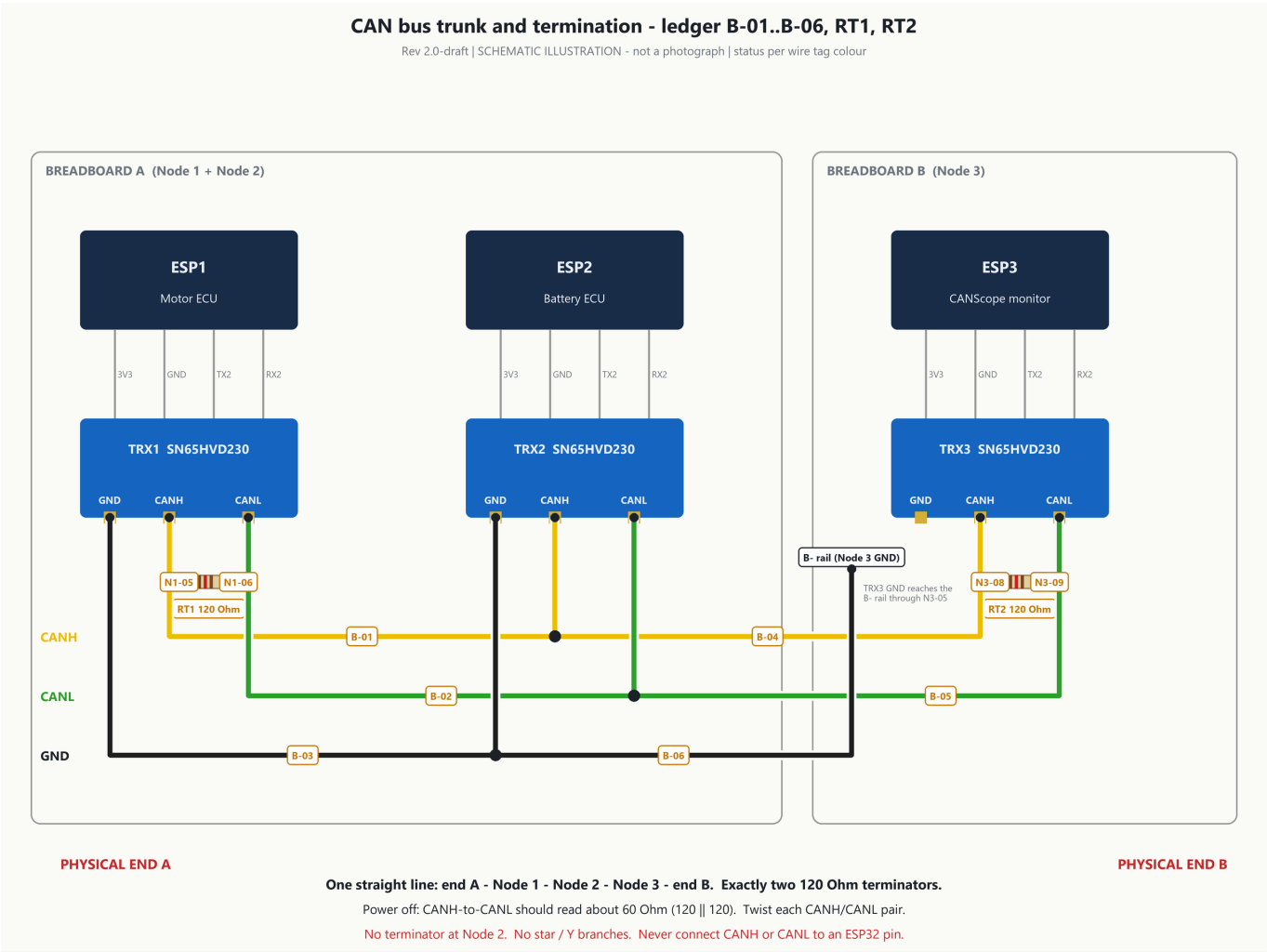
# 11. Reviewed transistor-driven buzzer circuit

Low-side NPN switch. GPIO13 -> R_B 1 kOhm -> Q1 base (Ib about 2.6 mA); R_PD 10 kOhm base -> GND keeps Q1 off while GPIO13 floats during reset; Q1 emitter -> GND; Q1 collector -> buzzer -; buzzer + -> Node 3 VIN (USB 5 V, measured 4.4-5.25 V at gate G12). For a buzzer of 30 mA or less the forced gain is about 12, which saturates the example parts. A 1N4148 across the buzzer (band toward +) is fitted only for an electromagnetic buzzer. The GPIO never touches the buzzer. D12 (GPIO12), next to D13, is a strapping pin and stays empty.

**BLOCKED:** Q1's leg order differs between parts (PN2222A/2N3904/S8050 = E-B-C, BC547 = C-B-E, flat face toward you). N3-28 to N3-32 stay empty until the part number and its datasheet pinout are recorded (G10). N3-35 (VIN -> buzzer +) is installed last, power off, after G10, G11 and G12.

| ID | From | To | Signal | Wire | Status | Gates | Check (power off) |
|---|---|---|---|---|---|---|---|
| N3-27 | ESP3 D13 (GPIO13), VIN-side pin 3 from USB | R_B 1 | BUZZER_DRIVE (GPIO13) | grey | **PROVISIONAL** | G2, G13 | Continuity D13 (3rd pin from USB, VIN side) to one end of the 1 kOhm. The 4th pin, D12, is a strapping pin: make sure it stays EMPTY. D13 has NO path to the buzzer except through Q1. |
| N3-28 | R_B 2 | Q1 B | Q1 base via 1 kOhm | lead | **BLOCKED** | G10 | Blocked until Q1 part number and its E/B/C order are recorded. |
| N3-29 | R_PD 1 | Q1 B | base pull-down (boot-quiet) | lead | **BLOCKED** | G10 | Blocked until Q1 pinout is recorded. |
| N3-30 | R_PD 2 | Board B top - rail (GND) | GND | lead | **BLOCKED** | G10 | Blocked until Q1 pinout is recorded. |
| N3-31 | Q1 E | Board B top - rail (GND) | GND (emitter) | black | **BLOCKED** | G10 | Blocked until Q1 pinout is recorded. |
| N3-32 | Q1 C | BZ1 - | BUZZER_LOW (collector) | lead | **BLOCKED** | G10, G11 | Blocked until Q1 pinout and buzzer polarity are recorded. |
| N3-33 | D1 A | BZ1 - | flyback diode anode | lead | **PROVISIONAL** | G11 | Only if G11 says electromagnetic. Band (cathode) goes to the + side. |
| N3-34 | D1 K | BZ1 + | flyback diode cathode | lead | **PROVISIONAL** | G11 | Band toward buzzer +. |
| N3-35 | ESP3 VIN, VIN-side pin 1 from USB | BZ1 + | 5V_USB_N3 (buzzer supply) | orange | **PROVISIONAL** | G11, G12 | Install LAST, with power off, after G12. No continuity from this wire to any 3V3 pin or rail. |

# 12. Physical CANH/CANL trunk




| ID | From | To | Signal | Wire | Status | Gates | Check (power off) |
|---|---|---|---|---|---|---|---|
| B-01 | TRX1 CANH | TRX2 CANH | CANH trunk, segment 1 | yellow | **PROVISIONAL** | G4 | Continuity TRX1 CANH to TRX2 CANH; no continuity CANH to CANL other than through terminators. |
| B-02 | TRX1 CANL | TRX2 CANL | CANL trunk, segment 1 | green | **PROVISIONAL** | G4 | Twist with B-01. Continuity TRX1 CANL to TRX2 CANL. |
| B-03 | TRX1 GND | TRX2 GND | GND reference, segment 1 | black | **PROVISIONAL** | G4 | Continuity Node 1 GND to Node 2 GND. |
| B-04 | TRX2 CANH | TRX3 CANH | CANH trunk, segment 2 (board A to B) | yellow | **PROVISIONAL** | G4 | Continuity TRX2 CANH to TRX3 CANH. |
| B-05 | TRX2 CANL | TRX3 CANL | CANL trunk, segment 2 (board A to B) | green | **PROVISIONAL** | G4 | Twist with B-04. Continuity TRX2 CANL to TRX3 CANL. |
| B-06 | TRX2 GND | Board B bottom - rail (GND) | GND reference, segment 2 | black | **PROVISIONAL** | G4, G14 | Continuity Node 2 GND to Node 3 GND rail. |

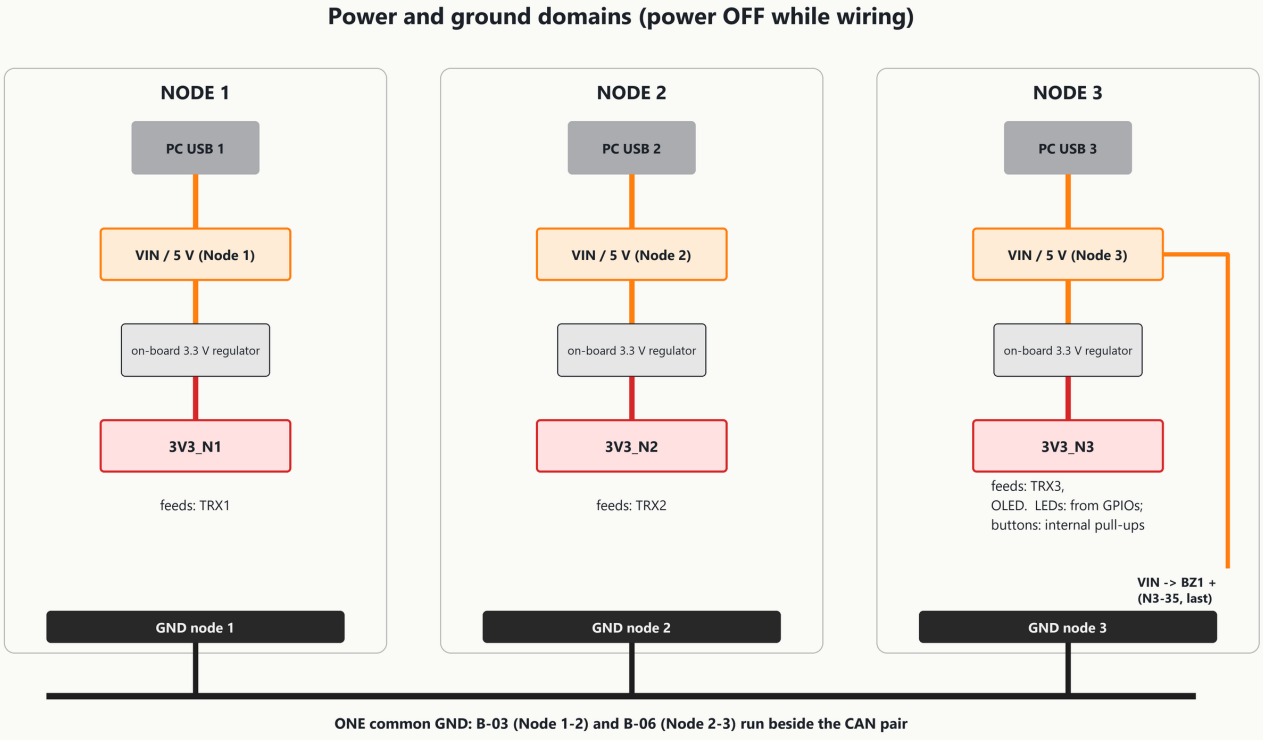
One linear bus: end A - Node 1 - Node 2 - Node 3 - end B. Each node taps the line directly at its transceiver pins; there are no Y branches. Twist each CANH/CANL pair and keep the board-to-board segment short. The GND wire runs beside the pair so all nodes share one reference.

# 13. Termination: exactly two, at the physical ends

| G6 reading per bare module | Action |
|---|---|
| All three > 10 kOhm (no onboard terminator) | Fit RT1 at TRX1 (N1-05/06) and RT2 at TRX3 (N3-08/09) |
| TRX1 or TRX3 reads about 120 Ohm | Omit that end's external resistor (the onboard one is the end terminator) |
| TRX2 reads about 120 Ohm | BLOCKED (row C-T2): its onboard resistor must be removed; send a photo first |

| ID | From | To | Signal | Wire | Status | Gates | Check (power off) |
|---|---|---|---|---|---|---|---|
| N1-05 | RT1 1 | TRX1 CANH | CANH (end A terminator) | lead | **PROVISIONAL** | G6, G13 | Fit only if G6 found no onboard terminator on TRX1. |
| N1-06 | RT1 2 | TRX1 CANL | CANL (end A terminator) | lead | **PROVISIONAL** | G6, G13 | RT1 measured 114-126 Ohm before fitting. |
| N3-08 | RT2 1 | TRX3 CANH | CANH (end B terminator) | lead | **PROVISIONAL** | G6, G13 | Fit only if G6 found no onboard terminator on TRX3. |
| N3-09 | RT2 2 | TRX3 CANL | CANL (end B terminator) | lead | **PROVISIONAL** | G6, G13 | RT2 measured 114-126 Ohm before fitting. |

# 14. Ground and power domains




**X NEVER join 3V3_N1, 3V3_N2 and 3V3_N3 together (two regulators fighting).**

**X NEVER join VIN/5 V of different nodes, and NEVER put 5 V on any 3V3 pin or rail.**

X No external 5 V supply is used. The buzzer takes 5 V from Node 3's own VIN pin while Node 3 is on USB.

X Unplug ALL three USB cables before any wiring change or any resistance / continuity test.

# 15. Unpowered continuity and resistance checks

All three USB cables unplugged for every check in this section. The PC's USB ground joins the nodes and would falsify readings. Meter leads: black in COM, red in V/Ohm. Never use the current jack.

| Check | Probe points | Pass | If not |
|---|---|---|---|
| Ledger walk-through | Each row of section 19, both ends | Every PROVISIONAL row's gates recorded; BLOCKED rows empty | Fix with power off |
| Supply shorts | 3V3 to GND on each node; Node 3 VIN to GND | No steady near-0 Ohm (may drift up as capacitors charge) | Find the short |
| 3V3 isolation | Node 1 3V3 to Node 2 3V3; any VIN to any 3V3 | Not near 0 Ohm | A rail or jumper joins them: remove |
| CANH continuity | TRX1 CANH - TRX2 CANH - TRX3 CANH | About 0 Ohm | Broken trunk |
| CANL continuity | TRX1 CANL - TRX2 CANL - TRX3 CANL | About 0 Ohm | Broken trunk |
| Polarity / isolation | CANH or CANL to GND, 3V3, VIN, any ESP32 pin | No continuity | Mis-wire: fix before power |
| Termination | CANH to CANL anywhere | About 60 Ohm (57-63) | 120: one missing; 40: three; open: broken; 0: short |
| Ground | GND of Node 1, 2, 3 to each other | About 0 Ohm | B-03 / B-06 missing |
| Buzzer isolation | D13 to BZ1 + and - | No continuity (only via Q1) | Remove the direct path |
| Strapping pin | D12 (GPIO12) | Nothing inserted | Remove |

# 16. First power-up sequence

- The bus stays isolated from any vehicle or external network.

- Plug in Node 1 only. Serial monitor at 115200 baud: banner lines start with '#'; expect '# CAN init: OK - controller running'. Measure TRX1 3V3-GND: 3.0-3.6 V. Watch for heat, smell, resets. Unplug.

- Repeat for Node 2 (it will report transmit errors alone: no node is powered to ACK; expected).

- Repeat for Node 3, and also measure VIN-GND (gate G12, 4.4-5.25 V). Unplug.

- Unpowered SN65HVD230 nodes do not disturb the bus (TI SLOS346O section 10.3), so the trunk stays wired.

- N3-35 (buzzer supply) is NOT installed while Q1 is BLOCKED. When unblocked, install it with all power off, after G10, G11 and G12.

- Never change a wire while any USB cable is plugged in. Unplug at once on heat, smell or repeated resets.

# 17. One-, two- and three-node bring-up

| Stage | Powered | Evidence that counts | Not evidence |
|---|---|---|---|
| 1 node | One at a time | Banner shows the expected pins (GPIO17/16), 500 kbit/s, CAN init OK; TRX 3V3 in range | Upload succeeded |
| 2 nodes | Node 1 + Node 2 | Node 1 prints JSON lines with "type":"frame" and canId 0x201, 0x202, 0x203 (~1 s) and 0x210 (~0.5 s); error counters stay low; no '# ALERT: BUS-OFF' | Node 2 printing 'Sent CAN ID' (only queued) |
| 3 nodes | all three | Node 3 (monitor, listen-only) prints the same IDs; terminators unchanged | OLED/LED/button activity (no firmware yet) |

The legacy test IDs carry fixed, unverified values (0x201 = 480, 0x202 = 120, 0x203 = 85, 0x210 = 1). They prove frames move; they are not motor or battery signals. Record the actual output in TASK.md and captures/.

## Firmware support today

- No firmware drives the OLED, LEDs, buttons or buzzer (GPIO21/22/25/26/27/32/33/18 are not referenced anywhere in firmware/).

- No motor (RPM/temperature) or battery (voltage/current/SoC) simulation and no fault injection exist in the current node1/node2 targets; they only start CAN with TX disabled.

- Only sketch_node2_test (PlatformIO env node2/legacy_sketch) transmits: fixed test frames 0x201, 0x202, 0x203 every 1 s and heartbeat 0x210 every 500 ms, UNVERIFIED units.

- firmware/node3 refuses to start CAN by design. Node 3 hardware (the CANScope monitor) must run the firmware/monitor target, which is listen-only on GPIO17/16 at 500 kbit/s.

# 18. Common wiring mistakes and troubleshooting

| Symptom | Likely cause | Safe response (power off first) |
|---|---|---|
| No frames at Node 1 | TX/RX swapped (CTX must get TX2); Rs high (standby); CANH/CANL swapped at one node; missing GND; wrong firmware | Check N1-03/04, N2-03/04, G5, polarity |
| Node 2 errors persist with Node 1 powered | Node 1 not ACKing: Rs standby, RX/TX mis-wired, bit rate mismatch | Compare banners; continuity |
| Bus reads ~40 Ohm | Third terminator (onboard) | Repeat G6 per module |
| Bus reads ~120 Ohm | One terminator missing | Check RT1/RT2 or onboard parts |
| Board will not boot | Something on D12 (strapping) or a short | Remove D12 wire; check rails |
| Button always pressed | Legs on the same contact | Use diagonal legs, re-check G8 |
| LED never lights | Reversed LED, or no firmware drives it yet | Diode test; note firmware gap |
| OLED blank | No firmware yet; or VCC/GND swapped by position | Re-read labels (G7) |
| Hot part / smell | Short, reversed supply, 5 V on 3V3 | Unplug everything; inspect before reuse |

# 19. Final wire-by-wire sign-off

Fill in on paper with all power off. A BLOCKED row must be physically empty. The electronic version, with coverage columns, is `validation/wire_coverage_checklist.md`.

| ID | From -> To | Status | Gates passed | Ends + meter | Init. |
|---|---|---|---|---|---|
| N1-01 | ESP1 3V3, 3V3-side pin 1 from USB -> TRX1 3V3 | PROVISIONAL | [ ] | [ ] | |
| N1-02 | ESP1 GND, 3V3-side pin 2 from USB -> TRX1 GND | PROVISIONAL | [ ] | [ ] | |
| N1-03 | ESP1 TX2 (GPIO17), 3V3-side pin 7 from USB -> TRX1 CTX | PROVISIONAL | [ ] | [ ] | |
| N1-04 | ESP1 RX2 (GPIO16), 3V3-side pin 6 from USB -> TRX1 CRX | PROVISIONAL | [ ] | [ ] | |
| N1-05 | RT1 1 -> TRX1 CANH | PROVISIONAL | [ ] | [ ] | |
| N1-06 | RT1 2 -> TRX1 CANL | PROVISIONAL | [ ] | [ ] | |
| N2-01 | ESP2 3V3, 3V3-side pin 1 from USB -> TRX2 3V3 | PROVISIONAL | [ ] | [ ] | |
| N2-02 | ESP2 GND, 3V3-side pin 2 from USB -> TRX2 GND | PROVISIONAL | [ ] | [ ] | |
| N2-03 | ESP2 TX2 (GPIO17), 3V3-side pin 7 from USB -> TRX2 CTX | PROVISIONAL | [ ] | [ ] | |
| N2-04 | ESP2 RX2 (GPIO16), 3V3-side pin 6 from USB -> TRX2 CRX | PROVISIONAL | [ ] | [ ] | |
| N3-01 | ESP3 3V3, 3V3-side pin 1 from USB -> Board B bottom + rail (3V3_N3) | PROVISIONAL | [ ] | [ ] | |
| N3-02 | ESP3 GND, 3V3-side pin 2 from USB -> Board B bottom - rail (GND) | PROVISIONAL | [ ] | [ ] | |
| N3-03 | Board B bottom - rail (GND) -> Board B top - rail (GND) | PROVISIONAL | [ ] | [ ] | |
| N3-04 | Board B bottom + rail (3V3_N3) -> TRX3 3V3 | PROVISIONAL | [ ] | [ ] | |
| N3-05 | Board B bottom - rail (GND) -> TRX3 GND | PROVISIONAL | [ ] | [ ] | |
| N3-06 | ESP3 TX2 (GPIO17), 3V3-side pin 7 from USB -> TRX3 CTX | PROVISIONAL | [ ] | [ ] | |
| N3-07 | ESP3 RX2 (GPIO16), 3V3-side pin 6 from USB -> TRX3 CRX | PROVISIONAL | [ ] | [ ] | |
| N3-08 | RT2 1 -> TRX3 CANH | PROVISIONAL | [ ] | [ ] | |
| N3-09 | RT2 2 -> TRX3 CANL | PROVISIONAL | [ ] | [ ] | |
| N3-10 | Board B bottom + rail (3V3_N3) -> OLED VCC | PROVISIONAL | [ ] | [ ] | |
| N3-11 | Board B bottom - rail (GND) -> OLED GND | PROVISIONAL | [ ] | [ ] | |
| N3-12 | ESP3 D21 (GPIO21), 3V3-side pin 11 from USB -> OLED SDA | PROVISIONAL | [ ] | [ ] | |
| N3-13 | ESP3 D22 (GPIO22), 3V3-side pin 14 from USB -> OLED SCL | PROVISIONAL | [ ] | [ ] | |
| N3-14 | ESP3 D25 (GPIO25), VIN-side pin 8 from USB -> R_G 1 | PROVISIONAL | [ ] | [ ] | |
| N3-15 | R_G 2 -> LED_G A | PROVISIONAL | [ ] | [ ] | |
| N3-16 | LED_G K -> Board B top - rail (GND) | PROVISIONAL | [ ] | [ ] | |
| N3-17 | ESP3 D26 (GPIO26), VIN-side pin 7 from USB -> R_Y 1 | PROVISIONAL | [ ] | [ ] | |
| N3-18 | R_Y 2 -> LED_Y A | PROVISIONAL | [ ] | [ ] | |
| N3-19 | LED_Y K -> Board B top - rail (GND) | PROVISIONAL | [ ] | [ ] | |
| N3-20 | ESP3 D27 (GPIO27), VIN-side pin 6 from USB -> R_R 1 | PROVISIONAL | [ ] | [ ] | |
| N3-21 | R_R 2 -> LED_R A | PROVISIONAL | [ ] | [ ] | |
| N3-22 | LED_R K -> Board B top - rail (GND) | PROVISIONAL | [ ] | [ ] | |
| N3-23 | ESP3 D32 (GPIO32), VIN-side pin 10 from USB -> BTN_A L1 | PROVISIONAL | [ ] | [ ] | |
| N3-24 | BTN_A L3 -> Board B top - rail (GND) | PROVISIONAL | [ ] | [ ] | |
| N3-25 | ESP3 D33 (GPIO33), VIN-side pin 9 from USB -> BTN_B L1 | PROVISIONAL | [ ] | [ ] | |
| N3-26 | BTN_B L3 -> Board B top - rail (GND) | PROVISIONAL | [ ] | [ ] | |
| N3-27 | ESP3 D13 (GPIO13), VIN-side pin 3 from USB -> R_B 1 | PROVISIONAL | [ ] | [ ] | |
| N3-28 | R_B 2 -> Q1 B | BLOCKED | n/a | [ ] empty | |
| N3-29 | R_PD 1 -> Q1 B | BLOCKED | n/a | [ ] empty | |
| N3-30 | R_PD 2 -> Board B top - rail (GND) | BLOCKED | n/a | [ ] empty | |
| N3-31 | Q1 E -> Board B top - rail (GND) | BLOCKED | n/a | [ ] empty | |
| N3-32 | Q1 C -> BZ1 - | BLOCKED | n/a | [ ] empty | |
| N3-33 | D1 A -> BZ1 - | PROVISIONAL | [ ] | [ ] | |
| N3-34 | D1 K -> BZ1 + | PROVISIONAL | [ ] | [ ] | |
| N3-35 | ESP3 VIN, VIN-side pin 1 from USB -> BZ1 + | PROVISIONAL | [ ] | [ ] | |
| B-01 | TRX1 CANH -> TRX2 CANH | PROVISIONAL | [ ] | [ ] | |
| B-02 | TRX1 CANL -> TRX2 CANL | PROVISIONAL | [ ] | [ ] | |
| B-03 | TRX1 GND -> TRX2 GND | PROVISIONAL | [ ] | [ ] | |
| B-04 | TRX2 CANH -> TRX3 CANH | PROVISIONAL | [ ] | [ ] | |

| ID | From -> To | Status | Gates passed | Ends + meter | Init. |
|---|---|---|---|---|---|
| B-05 | TRX2 CANL -> TRX3 CANL | PROVISIONAL | [ ] | [ ] | |
| B-06 | TRX2 GND -> Board B bottom - rail (GND) | PROVISIONAL | [ ] | [ ] | |

| Record | Value |
|---|---|
| ESP32 shield marking N1 / N2 / N3 (G1) | |
| Transceiver IC marking and labels N1 / N2 / N3 (G4) | |
| Rs state per module (G5) | |
| Onboard CANH-CANL reading per module (G6) | |
| OLED label order and address (G7) | |
| Transistor part number and E-B-C (G10) | |
| Buzzer type / rating / current (G11) | |
| Node 3 VIN voltage (G12) | |
| Bus CANH-CANL resistance, unpowered | |
| Two-node result (IDs seen at Node 1) | |
| Three-node result (IDs seen at Node 3) | |
| Photos saved at | |
| Checked by / date | |

## Evidence references

**E-TI**: Texas Instruments SN65HVD23x datasheet SLOS346O (rev. April 2018): pin 1 D (TXD input), 2 GND, 3 VCC (3.0-3.6 V recommended, 6 V abs max), 4 R (RXD output), 5 Vref, 6 CANL, 7 CANH, 8 Rs. Rs strong pull-down = high-speed, 10-100 kOhm to GND = slope control, >=0.75 VCC = standby (driver OFF, receiver listen-only). D/R abs max VCC + 0.5 V. ICC 10 mA typ / 17 mA max. Package marking VP230.
**E-ESP32**: Espressif ESP32 Series datasheet: strapping pins GPIO0, GPIO2, GPIO5, MTDI (GPIO12), MTDO (GPIO15); GPIO34-39 input-only with no internal pull-ups; internal pull-up about 45 kOhm.
**E-WROOM**: Espressif ESP32-WROOM-32 datasheet v3.8: GPIO6-11 connected to the integrated SPI flash (do not use); IO16 (module pin 27) and IO17 (module pin 28) are ordinary GPIOs on WROOM-32 (they are PSRAM pins only on WROVER modules).
**E-ARD**: framework-arduinoespressif32, variants/esp32/pins_arduino.h (installed in C:/canscope-pio): SDA = 21, SCL = 22 (Wire default pins for board esp32dev); TX = 1, RX = 3 (USB serial UART0).
**E-FW**: Repository firmware: firmware/node1/include/node1_hw_config.h, firmware/node2/include/node2_hw_config.h and firmware/monitor/include/monitor_hw_config.h set CAN TX GPIO17, RX GPIO16, 500 kbit/s; sketch_node1_test.ino and sketch_node2_test.ino use the same. firmware/node3 refuses to start CAN (all values UNVERIFIED).
**E-DOIT**: Widely published DOIT ESP32 DevKit V1 30-pin silkscreen order. This is NOT a manufacturer schematic: clone boards vary, so gate G2 compares it with the builder's board.
**E-CALC**: Hand calculation recorded in validation/correction_register.md (LED current, base drive, termination).